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PreAuth Copilot

Getting patients a yes in minutes, not days: an AI copilot for US prior authorization and India's cashless pre-authorization

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How I would build this. Not something I shipped.

Most prior auth delays are not hard medical calls. They are incomplete files. So I would build AI that makes every request complete the first time, and I would never let the AI say no.

THE SHORT VERSION

The problem. US physicians handle 40 prior auths a week each and spend 13 hours on them.[3] In India, 6 in 10 insured patients surveyed waited 6 to 48 hours to get approved and discharged.[19] Regulators in both countries have now put a clock on the payer.[1,16]

My bet. Start on the provider side. The copilot reads the chart, matches it to the payer's rules, cites the evidence and flags gaps before anything is submitted. A human signs off. The AI never denies.

US vs India. In the US I sell time back to specialty practices and ride the 2027 FHIR prior auth APIs. In India I sell speed to TPAs and insurers first: 17 TPAs settle 69% of health claims,[17] and the insurer carries the 3-hour discharge clock.

How I'd know it works. North star: first-time-right authorizations. Hard guardrail: no rise in denials for any patient group.

01 Why this problem, and why me

At ICICI Lombard I have spent two and a half years putting AI into motor insurance: vision-based vehicle inspection, OCR on issuance documents, ML fraud models in claims and a claim admissibility engine with 20+ automated checks. Prior authorization is the same shape of problem in healthcare. A request arrives with messy documents, someone checks it against policy rules, and the customer waits.

What pulled me in is that the wait is now regulated in both markets. CMS requires Medicare Advantage, Medicaid and CHIP plans to decide within 72 hours for urgent requests and 7 days for standard ones, with a specific reason for every denial.[1] IRDAI requires Indian insurers to decide a cashless request within 1 hour and give final authorization at discharge within 3 hours.[16] When a regulator puts a clock on a workflow, the product that makes the clock easy to meet gets bought.

02 The problem in two markets

United States: the practice does the proving

Prior auth is the payer saying "show us this is needed before we pay." The physician's office does the showing. In the AMA's 2025 survey, physicians reported 40 prior auths a week, 13 hours of physician and staff time spent on them, and 40% have staff who do nothing else. Over a quarter (26%) said prior auth had led to a serious adverse event for a patient.[3] MGMA found 60% of practices need at least three employees to finish one request.[7]

The denial data is telling. Medicare Advantage plans made about 53 million prior auth decisions in 2024 and denied 7.7%. Only 11.5% of denials were appealed, but 80.7% of those appeals were partly or fully overturned.[4] Some of that is real clinical disagreement. A lot of it, I would bet, is a first request that lacked the right evidence. The plans agree complete files get fast answers: their 2025 pledge targets real-time answers in 2027 for 80% of electronic approvals that arrive with all needed documentation.[5]

India: the pain is at discharge

India's version is cashless pre-authorization. The hospital's insurance desk sends a request to the insurer or its TPA at admission, then the final bill at discharge. Insurers settled 3.26 crore health claims in FY25; 58% by count were cashless and 69% went through TPAs.[17] The pain peaks at discharge: in a LocalCircles survey, 6 in 10 policyholders who claimed said it took 6 to 48 hours to get approved and discharged.[19] The family waits in the lobby, the bed stays blocked, and the desk chases the TPA by email and phone.

Table 1. Same problem, two very different markets

	United States	India
Who waits	The patient, for a scan, surgery or drug to be scheduled	The patient and family, at admission and worst of all at discharge
The clock	72 h urgent, 7 days standard, from 2026 (CMS-0057-F)[1]	1 h for cashless, 3 h for final authorization (IRDAI, 2024)[16]
Who decides	Health plan UM teams, often run by UM vendors	The insurer, or one of 17 licensed TPAs[17]
Rails	Portals, fax, phone. ePA adoption 40%.[6] Payer Prior Auth APIs due Jan 2027[1]	Email, TPA portals, phone. NHCX gateway live with 34 insurers and TPAs (2024),[20] not mandatory for hospitals[21]
Documents	EHR notes, CPT and ICD-10 codes, mostly typed	Scanned and handwritten notes, mixed languages, package rates
Economics	High value per request, labour-heavy practices	Low value per request, huge volume, insurer pays for delay

03 Who I am building for

Composites built from the research above, to be validated in discovery.

Persona	Their day	What they need
Maria, prior auth coordinator, 14-doctor ortho group, Ohio	Six payer portals, re-keying chart notes, fax and phone queues	A complete packet in one click, and a warning before submit
Dr. Shah, orthopaedic surgeon, same group	Evenings on peer-to-peer calls for MRIs he knows are needed	Fewer denials; appeal letters he only checks and signs
Ravi, insurance desk, 250-bed hospital, Hyderabad	Pre-auth forms from handwritten notes, TPA queries by email, families at 9 pm	Fewer queries and a live status he can show the family
Dr. Anitha, medical officer at a TPA	Hundreds of requests a day against policy wording and package rates	Clean, pre-checked files so the easy ones go fast

04 How big is this

Table 2. Market sizing. Rows marked estimate are my own maths, not published figures

Measure	Number	Source
US active physicians in direct patient care, 2024	866,460	AAMC[8]
Physician and staff hours on prior auth per physician per week	13	AMA 2025[3]
Practice time spent on prior auth per year	~270M hours, ~$8B	Estimate: 866k × 13 h × 48 weeks, halved for AMA sample bias, at $30/h loaded
Savings CMS expects from its rule	~$15B over 10 years	CMS[2]
India health claims settled, FY25	3.26 crore (₹94,248 cr)	IRDAI[17]
Share settled cashless, by count	58%	IRDAI[17]
Cashless authorization decisions per year	~3.8 crore	Estimate: 3.26 cr × 58% × 2 (pre-auth and final)
Health insurance premiums, FY25	> ₹1.2 lakh crore, ~9% growth	Ministry of Finance[18]

What the numbers tell me. The US is a labour market: at $150 per physician per month across ~300,000 physicians in prior-auth-heavy specialties, the serviceable market is about $540M a year (both inputs are my estimates). India's revenue per request is far smaller: at ₹50 per authorization the pool is roughly ₹190 crore a year (estimate). So India is

a speed and volume game, and the buyer is whoever pays for the delay. Money is already moving: Menlo Ventures reported AI spend on prior auth grew about 10x in 2025.[15]

05 Why AI here, and where not

Where AI earns its place: reading unstructured evidence (notes, imaging reports, scanned discharge summaries), matching it to criteria written in prose like "failed six weeks of conservative therapy," drafting query replies, appeals and family updates in Telugu or Hindi, and predicting which requests will get pended.

Where I would not use AI

- **Eligibility, benefits, code validity and package-rate maths.** These are lookups. Rules are cheaper, faster and auditable.
- **The decision to deny.** That stays with a licensed clinician, always. California's SB 1120 requires physician oversight of AI-assisted utilization review,[10] CMS has told Medicare Advantage plans that algorithms can assist but each decision must rest on the individual patient,[11] and UnitedHealth faces a lawsuit alleging an algorithm drove denials that were mostly reversed on appeal.[12] Six in 10 physicians already worry AI will push denials up.[3] A copilot that can say no loses the provider's trust on day one.

My rule of thumb from claims work: automate the checks before you automate the judgment.

06 The product

In one line: PreAuth Copilot sits inside the provider's workflow, turns a chart into a complete, evidence-cited prior auth packet, and keeps it moving until there is a decision.

1. **Build the packet.** From an order (US) or admission note (India), it pulls patient, plan, codes and clinical evidence into that payer's form.
2. **Gap check.** Before submit, it compares the packet with the payer's criteria, shows what is missing and gives a readiness score. This is the feature I would bet the product on.
3. **Submit and track.** FHIR API where the payer has one, NHCX in India, portal automation as fallback. One status view across payers.
4. **Answer queries and appeals.** It drafts the response from the chart and cites the pages.
5. **Discharge fast lane (India).** It checks the final bill against the approved amount, sub-limits and package rates so final authorization goes out clean inside 3 hours, and keeps the family updated on WhatsApp.

Left out of v1 on purpose: payer-side auto-approval, pharmacy prior auth (different rails, already crowded) and a US patient app. Each would slow down proving the core loop.



Figure 1. How a request moves through PreAuth Copilot. AI reads, matches and drafts. People approve, submit and decide.

07 The AI approach

Build vs buy

- **Buy the foundation model:** a frontier LLM through a HIPAA-eligible cloud with a BAA in the US, and an India-region deployment for India. Training our own would be wasted money.
- **Buy OCR, but test it on our own files.** From my work on OCR-based issuance, vendor benchmarks rarely match real scans, so I would bake off two or three vendors on Indian hospital documents.
- **Build what compounds:** the payer policy library (criteria versioned by effective date), the evidence-citation layer, the gap-check model and the eval harness. The model is a commodity; the policy library and outcome data are the moat. Later, a small fine-tuned open-weights model handles high-volume extraction in India, where money per request is thin.

Data I need

Payer policies and documentation rules, which the CMS Prior Auth API has to expose from 2027;[1] 2,000 to 5,000 past requests per specialty with outcomes and query reasons from

design partners; in India, policy wordings, sub-limits, package rates and TPA query history. Licensed criteria like InterQual or MCG only through a payer partner.

Human in the loop

In v1 a coordinator reviews every packet. Every AI statement links to its source line; if it cannot cite, it cannot say it. Auto-submit unlocks per payer and procedure only after four weeks of 98%+ packets accepted unedited (my target). Nothing adverse is ever automated.

Table 3. Evals and the bar to ship. Bars are my starting targets, to be recalibrated in shadow mode

What I test	How	Bar
Critical fields (member ID, codes, dates, amounts)	500 labelled cases per specialty, exact match	≥ 98%
Evidence citation precision	Clinicians check each cited line really supports the criterion	≥ 95%
Unsupported statements	Automated span check plus weekly human audit	< 0.5% of packets
Gap-check recall	Backtest: did we flag what caused real queries?	≥ 80% in v1
Fairness	Readiness score and outcomes by age, sex, language, region	No group > 2 pts worse
Latency and cost	p95 time to draft; model cost per authorization	< 90 s; < $0.40 US, < ₹8 India

Table 4. Failure modes and guardrails

What could go wrong	What I build in
The copilot invents evidence	Every statement needs a citation; a failed span check blocks the packet
The payer changed its policy last week	Policies versioned by effective date; alert when query rates drift
Coordinators rubber-stamp	Weekly sampled audits; alarm if edits fall to near zero while errors rise
Gaming toward codes that get approved	It only cites what is in the chart and never suggests diagnoses
Data leaks, or poor accuracy on Telugu, Hindi or handwriting	BAA, no training on customer data, India data stays in India; low-confidence fields go to a human, accuracy tracked by language

Responsible AI and regulation

US: HIPAA, CMS-0057-F,[1] CMS's position on algorithms in Medicare Advantage[11] and state laws like SB 1120.[10] CMS is itself testing AI-assisted prior auth (the WISeR model, six states, from 2026) with licensed clinicians making every non-payment call.[9] I would hold the same

line. **India:** IRDAI's timelines,[16] the DPDP Act 2023 and its Rules on consent, and ABDM consent flows for NHCX.

08 How I would measure success

North star: first-time-right authorizations, the share of requests approved with no query or pend, inside the payer's clock. The patient feels it (care on time), the coordinator feels it (no rework) and so does the payer (less review). It only moves if the copilot is accurate and actually used.

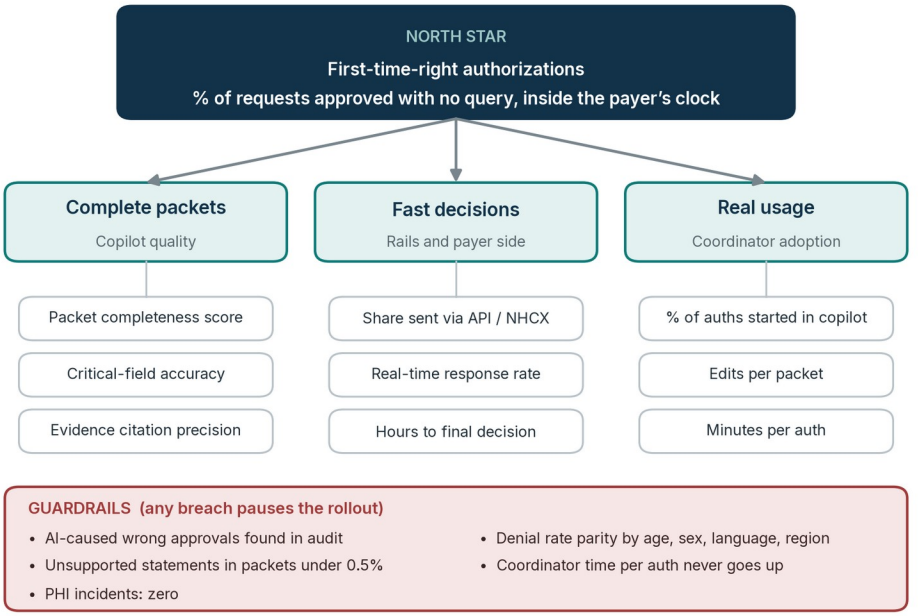


Figure 2. Metrics tree. Three input metrics feed the north star; guardrails can stop a rollout on their own.

09 MVP and roadmap

Phase	Scope	Exit criteria
0. Discovery Weeks 0–6	15 interviews per market; map 200 real requests. US: imaging and ortho surgery. India: five planned procedures, three hospitals, one TPA	Baseline time per auth, query rate, turnaround
1. Shadow MVP Months 2–4	Packet builder and gap check run beside the team. Humans still submit the old way	Every eval bar met on live traffic
2. Assisted Months 4–9	Submit from the copilot; tracking, query and appeal drafts; India discharge fast lane with WhatsApp updates	First-time-right +15 pts; minutes per auth −40%
3. Connected Months 9–18	Payer FHIR Prior Auth APIs (US, 2027) and NHCX (India). Real-time clean-lane pilot with one payer	50%+ of volume on APIs; zero guardrail breaches

10 Go-to-market: what I would do differently in each market

United States: sell time back to the practice

- **Buyer:** VP of Revenue Cycle at specialty groups (orthopaedics, imaging, oncology) and mid-size health systems; the prior auth lead is the champion. **Wedge:** one specialty, the top five payers in one state.
- **Pricing:** per physician per month, or per authorization for small groups, sold on staff hours saved.
- **Competition:** payer-side platforms such as Cohere Health (reports 85% real-time approvals)[13] and Availity AuthAI (reports near real-time approval recommendations on 75% of requests)[14] speed up the payer, and EHR vendors could bundle. My answer: stay EHR-agnostic and provider-loyal, and make ours the packets those payer engines approve fastest.

India: sell speed to whoever pays for the delay

- **Buyer:** TPAs and insurers first. They carry the 3-hour discharge clock, and the market is concentrated: 17 TPAs settle 69% of claims.[17] Large hospital chains second.
- **Pricing:** per authorization for TPAs and insurers; a low monthly fee per hospital desk via hospital software (HMIS) vendors. **Design for** scanned paper, low bandwidth, WhatsApp and local languages.
- **Competition:** Medi Assist, the largest TPA, launched an AI stack for cashless operations in November 2025,[22] and big insurers build in-house. My edge is sitting at the hospital desk where the file is created, and working with every TPA.

11 Risks and how I would de-risk them

Risk	Why it is real	How I de-risk
Seen as an AI denial tool	Physicians are primed for it[3]	It cannot output a denial; share fairness metrics
Payer APIs arrive late	ePA is still at 40%[6]	Portal fallback; start with payers already live
EHR vendor bundles it	They own distribution	Multi-EHR, specialty depth, outcome data
Indian hospitals do not adopt	Desks are stretched	Sell to TPAs, who pull hospitals in; WhatsApp-first
Thin India margins	₹ per request is small	Small models, batching, price on penalties avoided

12 What I would measure in the first 90 days

Days 1–30, baseline. Today's minutes per auth, query rate, first-time-right rate and turnaround for two US specialties and one TPA. Label 500 cases. Sign three design partners per market.

Days 31–60, shadow. Run silently on live traffic. Track field accuracy, citation precision and gap-check recall against real queries. Fix or cut anything under the bar.

Days 61–90, assisted pilot with about ten coordinators. Targets: 60%+ of auths started in the copilot, minutes per auth down 40%, first-time-right up against a control group, zero guardrail breaches. Paid rollout is a go or no-go on those numbers.

> If a coordinator in Ohio gets an hour of her day back, and a family in Hyderabad leaves the hospital before dinner instead of after midnight, this product sells itself. The AI is the easy part. Trust is the product.

Sources

Estimates and targets are labelled where they appear. Cohere Health, Availity and Medi Assist figures are the companies' own claims. Lawsuit details are allegations.

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